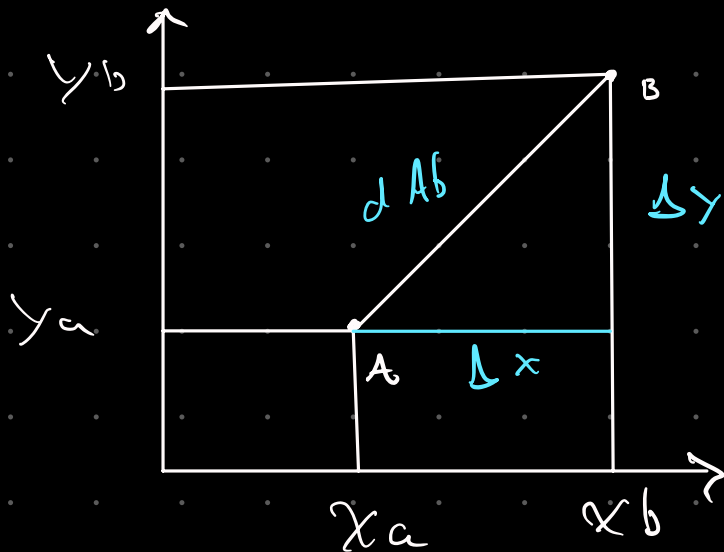


Marcação Geo. Anal.

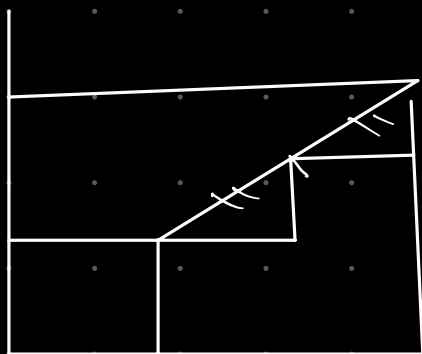
1) Distância entre dois pontos



$$\Delta y \quad (d_{AB})^2 = \Delta x^2 + \Delta y^2$$

$$d_{AB} = \sqrt{(x_b - x_a)^2 + (y_b - y_a)^2}$$

2) Ponto Médio



$$\frac{x_a + x_b}{2} = x_m$$

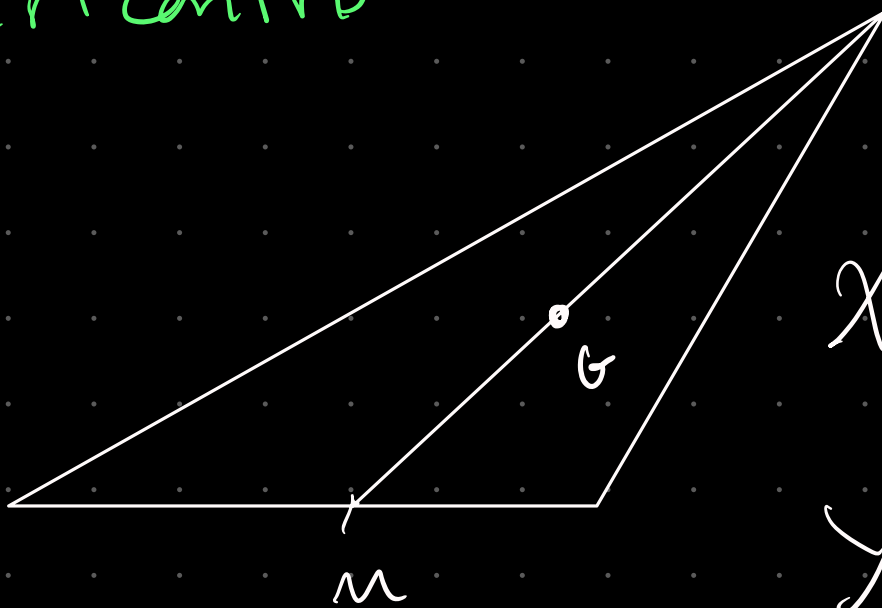
$$y_m = \frac{y_a + y_b}{2}$$

$$\left\{ \vec{M} = \frac{\vec{A} + \vec{B}}{2} \right\}$$

Exemplu AB é Diâmetro de
Circunferência $C = (3, 2)$ sabendo
que $A = (-1, 5)$ determinar B

$$B = (7, 1)$$

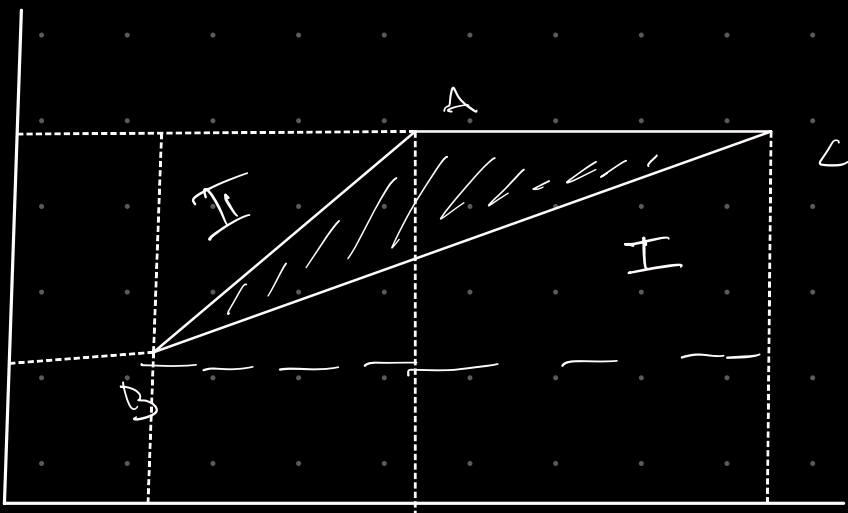
3) Baricentro



$$x_G = \frac{x_a + x_b + x_c}{3}$$

$$y_G = \frac{y_a + y_b + y_c}{3}$$

3) Area de polígonos



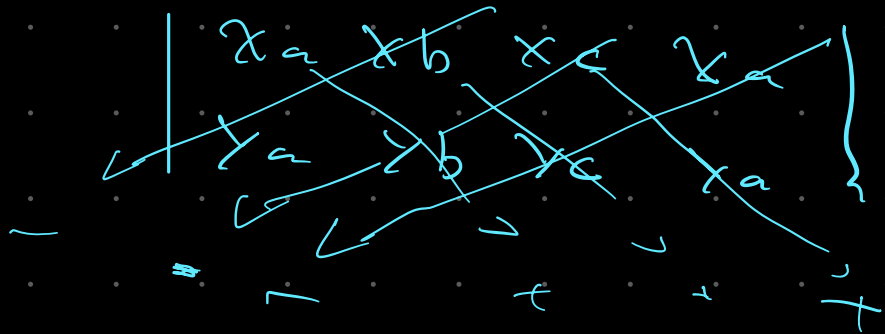
$$[ABC] = A_{ret} - (A_I + A_{II} + A_{III})$$

fazendo Distribuição

$$\frac{|x_a y_b + x_b y_c + x_c y_a - x_b y_a - x_c y_b - x_a y_c|}{2}$$

$$= \frac{|S|}{2} \Rightarrow S = \begin{vmatrix} x_a & y_a & 1 \\ x_b & y_b & 1 \\ x_c & y_c & 1 \end{vmatrix} \quad \text{Triângulo}$$

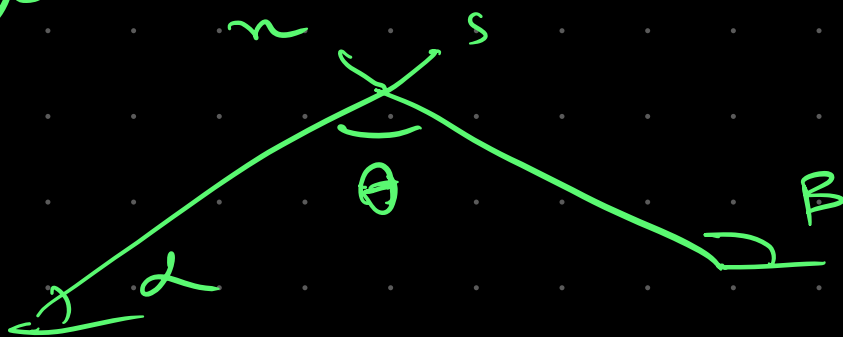
Operador area de couchy



Distancia de ponto a reta

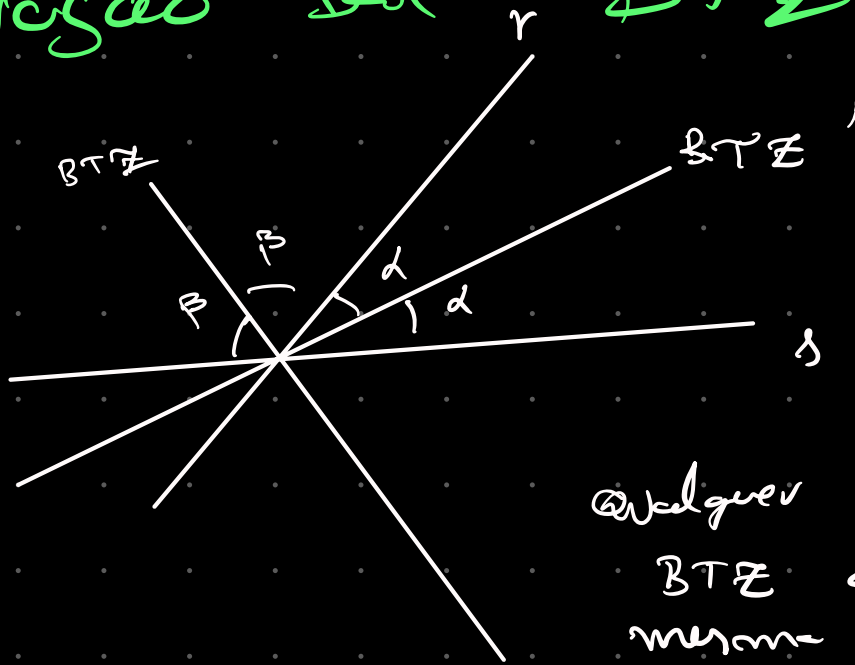
$$d_{p,r} = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$$

Angulo entre retas



$$\tan \theta = \left| \frac{m_s - m_r}{1 + m_s m_r} \right|$$

Equação da BTZ's



Qualquer ponto de
BTZ dista a
mesma distância
das retas

$$d_{p,r} = d_{p,s}$$

$$\frac{|ax + by + c|}{\sqrt{a^2 + b^2}} = \frac{|a'x + b'y + c'|}{\sqrt{a'^2 + b'^2}}$$